

- Core relies on its package manager to receive updates.
- 5. Microservices architecture:** A microservices architecture allows a mix of technologies across a service boundary. This technology mix enables a gradual embrace of .NET Core for new microservices that work with other microservices or services. For example, you can mix microservices or services developed with the .NET Framework, Java, Ruby, or other monolithic technologies. There are many infrastructure platforms available. Azure Service Fabric, for instance, is designed for large and complex microservice systems. Azure App Service is a good choice for stateless microservices. Other alternatives based on Docker fit any kind of microservices approach, as explained in the section on 'Containers'. All these platforms support .NET Core and make them ideal for hosting your microservices.
 - 6. Open source:** .NET Core is open source and uses the MIT and Apache 2 licences. It is part of the .NET Foundation, which is an independent non-profit supporting the innovative, commercial yet open source-friendly .NET ecosystem. Over 25,000 developers from over 1700 companies outside Microsoft are currently contributing to the .NET open source code base. The .NET community is growing fast, and already has a good number of community projects and libraries available for free. In addition to the community and Microsoft, the Technical Steering Group members like Google, JetBrains, Red Hat, Samsung and Unity are guiding the future of the .NET platform.
 - 7. Tools and productivity:** The Visual Studio product family provides a great .NET development experience on Windows, Linux, and macOS. The Visual Studio Marketplace has thousands of editor extensions from Microsoft and others. If you prefer to use a different editor, there are .NET command line tools and plugins for many popular editors. .NET helps you develop high quality applications faster. Modern language constructs like generics, Language Integrated Query (LINQ), and asynchronous programming make developers productive. Combined with the extensive class libraries, common APIs, multi-language support and the powerful tooling provided by the Visual Studio family, .NET is the most productive platform for developers.
 - 8. Command line tools:** These include easy-to-use command line tools for local development and in continuous-integration scenarios.
 - 9. Performance:** .NET Core is fast. Yes, faster than Node.js and Go. Really fast! That means applications provide better response times and require less compute power. Stackoverflow serves 5.3 million page views a day on just nine servers. The popular TechEmpower benchmark compares Web application frameworks with tasks like JSON serialisation, database access, and server-side template rendering, and .NET Core performs faster than any other popular framework.
Microsoft recommends running .NET Core with ASP.NET Core for the best performance and scale. This becomes important when hundreds of microservices are used, in which case fewer servers and virtual machines are needed. The efficiency and scalability gained can translate to a better user experience in addition to cost savings. Bing.com runs on .NET Core 2.1 and the platform has seen a 34 per cent improvement in its performance, which means fewer machines are required. New Zealand headquartered Raygun has reported that it increased its throughput 2000 per cent by switching from Node.js to .NET Core.
 - 10. Trusted and secure:** .NET Core provides you with immediate security benefits through its managed run time. A collection of services prevents critical issues like bad pointer manipulation or malicious attempts to alter compiled code. Microsoft takes security very seriously and releases updates quickly when threats are discovered.
 - 11. A vast ecosystem:** You can leverage the vast .NET ecosystem by incorporating libraries from the NuGet package manager, the extensive partner network, and the Visual Studio Marketplace. You can also find answers to technical challenges from the community, MVPs, and large support network.
 - 12. Support:** .NET Core is supported by Microsoft, on Windows, macOS and Linux. It is updated for security and quality several times a year. .NET Core binary distributions are built and tested on Microsoft-maintained servers in Azure and supported just like any Microsoft product. Red Hat supports .NET Core on Red Hat Enterprise Linux (RHEL). It builds .NET Core from source and makes it available in

Table 1: The .NET Core release table

Version	Announcement date	Cross-platform support	Production ready (GA)
.NET Core 1.0	June 27, 2016	Yes	Yes
.NET Core 2.0, ASP.NET Core 2.0, Entity Framework Core 2.0	August 14, 2017	Yes	Yes
.NET Core 2.1, ASP.NET Core 2.1, Entity Framework Core 2.1, IoT	May 30, 2018	Yes (including SignalR)	Yes
.NET Core 2.2 & 3.0 (new additions-WinForms, WPF, UWP)	May 7, 2018 (under preview)	Yes (except WinForms, WPF and UWP, which are Windows-only)	No